

Análisis Matemático de Algoritmo

Generado por Project Diophantus

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Resumen Ejecutivo

Este documento presenta la traducción completa de un algoritmo computacional a tres formas matemáticas distintas. Las ecuaciones resultantes, debido a la complejidad logarítmica del algoritmo original (Miller-Rabin), son extensas pero polinomialmente acotadas.

Part I

Función de Transición

$$x[t+1] = x$$

$$y[t+1] = 0$$

Part II

Conversión a Polinomio

Ecuación Unificada $P=0$

$$\begin{aligned} & (e_0 \cdot (1 - e_0))^2 + (e_0 \cdot ((1) - (n) - (e_1^2 \\ & + e_2^2 + e_3^2 + e_4^2)))^2 \\ & + ((1 - e_0) \cdot ((n) - (1) - 1 - (e_5^2 + e_6^2 + e_7^2 \\ & + e_8^2)))^2 + (e_{11} - (n - 1))^2 \\ & + (P_{factorial}(factorial, factorial, e_{11}, e_{10}))^2 \\ & + (e_9 - (n \cdot e_{10}))^2 + (RET - ((e_0) \cdot (1) \\ & + (1 - e_0) \cdot (e_9)))^2 + (x[t+1] - (x))^2 + (y[t+1] - (0))^2 = 0 \end{aligned}$$

Part III

Fórmulas Generadoras

$$\begin{aligned} G = & 1 - ((e_0 \cdot (1 - e_0))^2 + (e_0 \cdot ((1) - (n) - (e_1^2 \\ & + e_2^2 + e_3^2 + e_4^2)))^2 \\ & + ((1 - e_0) \cdot ((n) - (1) - 1 - (e_5^2 + e_6^2 + e_7^2 \end{aligned}$$

$$\begin{aligned}
& + e_8^2)))^2 + (e_{11} - (n - 1))^2 \\
& + (P_{factorial}(factorial, factorial, e_{11}, e_{10}))^2 \\
& + (e_9 - (n \cdot e_{10}))^2 + (RET - ((e_0) \cdot (1) \\
& + (1 - e_0) \cdot (e_9)))^2 + (x[t + 1] - (x))^2 + (y[t + 1] - (0))^2)
\end{aligned}$$

Part IV

Fórmulas Matemáticas (Estilo Willans)

1 Representación Simbólica

$$\begin{aligned}
f(N) &= \sum_{R=0}^{\infty} R \cdot \lfloor \frac{1}{1 + \mathcal{D}_{sym}} \rfloor \\
&\mathcal{D}_{sym} \\
&= (\epsilon_0 \\
&\mathit{mathitcdot}(1 - \epsilon_0))^2 + (\epsilon_0 \\
&\mathit{mathitcdot}((1) - (n) - (\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2)))^2 + \\
&((1 - \epsilon_0) \\
&\mathit{mathitcdot}((n) - (1) - 1 - (\epsilon_5^2 + \epsilon_6^2 + \epsilon_7^2 + \epsilon_8^2)))^2 + \\
&(\epsilon_{11} - (n - 1))^2 + (\Phi_{factorial}(factorial, factorial, \epsilon_{11}, \epsilon_{10}))^2 + \\
&(\epsilon_9 - (n \\
&\mathit{mathitcdot}\epsilon_{10}))^2 + (RET - ((\epsilon_0) \\
&\mathit{mathitcdot}(1) + (1 - \epsilon_0) \\
&\mathit{mathitcdot}(\epsilon_9)))^2 + \\
&(x - (x))^2 + (y - (0))^2
\end{aligned}$$

2 Representación Operacional

$$\begin{aligned} f(N) &= \sum_{R=0}^{\infty} R \cdot \left[\frac{1}{1 + \mathcal{D}(N, R, \mathbf{x})} \right] \\ &= \sum_{\mathbf{x} \in \mathbb{N}^{19}} \left((x_{134}(1 - x_{13}))^2 + (x_{134}((1) - (x_{19}) - (x_{16}^2 + x_7^2 + x_8^2 + x_{11}^2)))^2 \right. \\ &\quad \left. + ((1 - x_{13})_4((x_{19}) - (1) - 1 - (x_6^2 + x_{12}^2 + x_{15}^2 + x_9^2)))^2 + (x_5 - (x_{19} - 1))^2 + (x_1(x_2, x_2, x_5, x_3))^2 \right. \\ &\quad \left. + (x_{14} - (x_{194}x_3))^2 + (x_{10} - ((x_{13})_4(1) + (1 - x_{13})_4(x_{14})))^2 + (x_{17} - (x_{17}))^2 + (x_{18} - (0))^2 \right) \end{aligned}$$

Vector de estado: $\mathbf{x} \in \mathbb{R}^{19}$ (variables internas anonimizadas).

Part V

Sistema Dinámico (Recurrencia Infinita)

Sistema de recurrencia generado (Placeholder)